

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

5KeTe5hnCJ6uCDiq2v869Q8BLahwcXt36hGutxjAxfg4

Date: 5/1/2025

Compounds: Ecstasy, Fentanyl, Heroin, Hydrocodone

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Opioid-Empathogen Hybrid

Formula: C22H28N2O2

Weight: 352.47

Drug-likeness: 0.45

Activity:

Analgesic and psychoactive

Target Analysis

Mu Opioid Receptor

Binding Affinity: -11.5

Primary target for opiate-based analgesics with significant risk for dependency

Serotonin Transporter (SERT)

Binding Affinity: -9.4

Targeted in mood regulation, with potential for empathogenic effects

Toxicity Profile

Risk Level: High

Affected Systems: Brain, Liver, Respiratory System

Mitigation Suggestions:

- Dosage control
- Antagonist co-administration
- Continuous monitoring

Pharmacokinetics

Absorption:

Rapid via oral and parenteral routes

Distribution:

Widely distributed, crosses blood-brain barrier

Metabolism:

Hepatic, primarily via CYP3A4

Excretion:

Renal, with small biliary excretion

Half-life: 3-7 hours

Detailed Analysis

The hypothetical Opioid-Empathogen Hybrid compound is designed by integrating the analgesic and empathogenic properties of current opioid and recreational drugs. With a combined molecular structure designed to target both the mu-opioid receptor and serotonergic pathways, this compound aims to exploit the synergistic effects of its parent drug properties. It poses strong clinical potential in both therapeutic interventions for pain and possibly adjunctive psychotherapy with its empathy-boosting effects. However, the introduction of such a compound comes with significant safety concerns, particularly regarding addiction liability, neurotoxicity due to serotonergic overactivity, and life-threatening respiratory depression. Careful structural optimization and clinical evaluations are necessary to mitigate these risks while maximizing its therapeutic potential.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>